Aayush Kumar

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Education

National Institute of Technology, Kurukshetra **Nov 2021 – May 2025**

B.Tech in Computer Engineering

Experience

Airbus — Software Developer 1 **Nov 2025 – Present**

- Build and maintain a **C++/Qt/QML** iPad application for flight-operations workflows, used daily by **40,000 pilots**, with a focus on responsiveness, stability and predictable behaviour.
- Cut main-thread memory usage by **23%** by profiling with Xcode Instruments and tightening object and resource lifetimes on hot code paths.
- Designed **state-machine-driven** workflows that guarantee deterministic, traceable state transitions in **safety-critical** scenarios.
- Integrated a fine-tuned aviation-domain small language model into the app for context-aware operational queries.

Airbus — DevOps Intern **Jun 2025 – Nov 2025**

- Built a proof of concept for migrating on-premises infrastructure to **AWS**, evaluating scalability, deployment feasibility and operational requirements.
- Automated cloud workflows with **AWS Lambda** and **S3**, and configured **Route 53** DNS for service integration.
- Built **Docker**-based CI/CD pipelines plus monitoring scripts and APIs for deployment validation and log-based troubleshooting.

Turing — Software Developer **Nov 2024 – Jun 2025**

- Built **GitHub Actions** evaluation pipelines that scored LLM output on backend engineering, systems, competitive programming and debugging tasks.
- Built a **Dockerized reinforcement-learning evaluation environment** that compared model output against reference annotations for **Meta** and **Alibaba** projects.
- Worked with enterprise client teams to identify recurring model failure cases and debugging patterns, improving RLHF dataset workflows.

Projects

lob-engine | ***C++20, Google Benchmark, GoogleTest, ASan/UBSan, GitHub Actions*** **Source Code**

- Built a price-time-priority **limit order book and matching engine** with a pre-allocated order pool and intrusive per-level queues (no allocation per order, O(1) cancel).
- Replaced `std::map` price levels with flat arrays and a bitset of non-empty levels: **20.1M → 34.0M ops/s** on a 1M-op mixed workload; a sparse-book benchmark exposed and fixed a full-band scan.
- Verified with a **differential test** against a naive reference book (20,000 random steps), mutation-checked by **12 hand-planted bugs, all caught**; CI under ASan/UBSan.

Trace Studios | ***Next.js, Flask, Raspberry Pi, Docker, Supabase, Turso*** **Live**

- Built and launched a full-stack **photo-booth platform** with photo capture, digital sharing and Raspberry Pi thermal printing.
- Implemented a Flask printing pipeline and cloud-to-edge gallery sync, with **HMAC-signed uploads** and QR-based photo access.

Technical Skills

Languages: C++, C, Python, Go, Java, Bash

Systems: Linux, performance profiling, state machines, Docker, Kubernetes, CI/CD (GitHub Actions)

Cloud: AWS (Lambda, S3, Route 53), GCP, monitoring and automation

Frameworks: Qt/QML, Flask, REST APIs

Achievements

- **Certified Kubernetes Administrator (CKA)**; AWS Certified Cloud Practitioner.
- Solved 950+ data structures and algorithms problems (LeetCode 250+, GeeksforGeeks 700+).

Leadership

- **President, SUC Dance Crew:** led a 35-member team to 7 inter-college wins. **Co-organiser, TEDxNITKKR:** ran the event end to end for 300+ attendees and 8 speakers.